

Robabeh Bashiri

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SUMMARY

Data-driven Materials Data Scientist skilled in Python, SQL, Machine Learning, Predictive Modeling & scientific data analysis who comes from a PhD-trained materials science and electrochemistry background. An analytical and quality-focused professional specializing in materials performance and energy data, with a mission to enable data-driven decisions that improve material reliability, accelerate innovation, and support sustainable energy solutions by translating complex experimental datasets into actionable, predictive insights.

SKILLS

Data Analysis | Scientific Data Analysis | Python (Data Analysis) | SQL (Foundational) | Exploratory Data Analysis (EDA) | Data Visualization | Statistical Analysis | Technical Writing | Project Management | Materials Science & Characterization | AWS | Generative AI | Machine Learning | Time Series | Data Cleaning

TECH PROJECTS

Real-Estate Price Prediction with Machine Learning | 11/2025 | [GitHub](#) |

Implemented machine learning techniques to replace traditional expert-based approaches for real estate pricing.

- Trained an algorithm to automatically discover pricing patterns from data by using Python, Machine Learning algorithms, Data preprocessing, and cleaning.
- Built a Decision Tree classifier in scikit-learn, learning key pricing patterns to classify high- vs. low-priced homes.
- Tuned model hyperparameters (max_depth 1–5) in Python, increasing validation accuracy from ~85% to ~88%.
- Persisted the optimized model with joblib, enabling efficient reuse and reproducible deployment.

Beta Bank Customer Churn Prediction | 10/2025 – 12/2025 | [GitHub](#) |

Developed a machine learning model to predict customer churn for a retail bank, identifying customers at risk of leaving and analyzing the key factors driving churn to support retention strategies.

- Built and evaluated baseline classification models, analyzing class imbalance between retained and churned customers to document initial performance limitations.
- Improved model performance using imbalance-handling techniques, applying multiple strategies and hyperparameter tuning on training and validation sets to select the optimal model.
- Tested and reported final model results, interpreting performance metrics to quantify churn risk and highlight the most influential drivers of customer attrition.

TECHNICAL WORK EXPERIENCE

Photocatalytic Modeling & Performance Optimization Project | 09/2019 – 12/2023 |

Led a multidisciplinary computational–experimental research project to optimize photocatalytic materials for hydrogen generation through electronic structure modeling, performance analysis, and data-driven design.

- Secured **\$250K** in competitive national research funding aligned with energy and advanced materials priorities.
- Linked computational results with experimental data to optimize materials for **hydrogen evolution performance**.
- Led cross-functional computational and experimental teams, delivering projects on time through structured planning, risk management, and documentation.

Standalone Solar Power System Development | 09/2017 – 01/2020|

Led the design and development of a standalone solar-powered electricity system integrating photoelectrochemical (PEC) and dye-sensitized solar cell (DSC) technologies to enable renewable electricity generation and hydrogen production.

- Secured \$300K in competitive research funding, defining technical scope and performance objectives aligned with renewable energy and sustainability priorities.
- Coordinated prototype fabrication and performance evaluation, analyzing system efficiency and power output to assess feasibility for standalone solar-powered electricity generation.

EXPERIENCE

California State University, Chico | Lecturer in Chemistry & Researcher | 03/2022 – Present | Chico, CA

Skills Used: Data Analysis, Statistical Analysis, Scientific Computing, Chemical Instrumentation, SOP Development, Project Management

- Analyzed experimental and laboratory datasets, improving material performance evaluation and supporting data-driven decisions in electrochemical and materials research.
- Managed and trained multidisciplinary research teams (7+ researchers), improving experimental consistency and data reliability through structured training and protocol enforcement.
- Characterized materials using advanced instrumentation (XRD, SEM/EDX, FTIR, UV-Vis, BET), translating complex measurement data into actionable insights on structure–property relationships.

University of Gdansk | Research Scientist | 01/2021 – 12/2021 | Gdańsk, Poland

Skills Used: Statistical Data Analysis, Exploratory Data Analysis (EDA), Data Visualization, Scientific Computing, Materials Characterization, Project Management

- Analyzed and visualized spectroscopic and electrochemical datasets (XRD, FTIR, EIS, I–V), identifying structure–performance relationships to support data-driven catalyst design.
- Processed raw instrument data using NOVA software, generating clean, analysis-ready datasets through statistical analysis and curve fitting to ensure reproducible results.
- Applied data-driven insights to guide experimental decisions, contributing to a 15% improvement in material efficiency through iterative analysis and optimization.
- Translated analytical findings through visualizations, technical reports, and presentations, enabling a clear understanding for research teams and funding stakeholders.

Universiti Teknologi PETRONAS | Research & Development Project Manager | 01/2017 – 12/2020 | Malaysia (

Skills Used: Quantitative Analytics, Statistical Data Analysis, Data Management, Project Management, Data Modeling, Proposal Writing, Stakeholder Engagement

- Secured \$1M+ in external research funding, delivering sponsor-aligned proposals by translating complex scientific objectives into data-driven, competitive grant applications.
- Integrated electrochemical datasets with advanced materials characterization (XRD, HRTEM, SEM, XPS, FTIR, UV-Vis), modeling structure–property–performance relationships, and improving research outputs by ~20%.
- Translated technical results into executive-ready reports and presentations, supporting funding agencies, industry partners, and public dissemination through conferences and stakeholder briefings.

EDUCATION

TripleTen | Applied Data Science & AI/ML Training | 2025 – Present

Coursework: Python, SQL, Databases, Machine Learning, Data Processing, Data Cleaning, Data Visualization

Universiti Teknologi PETRONAS | Doctor of Philosophy (PhD), Chemical Engineering | 2011 – 2016

Research focus: Advanced catalytic materials, renewable energy systems, hydrogen production

Payame Noor University | Master of Science (MSc), Inorganic Chemistry | 2006 – 2008

Focus: Materials design, characterization, and nanostructured coordination polymers